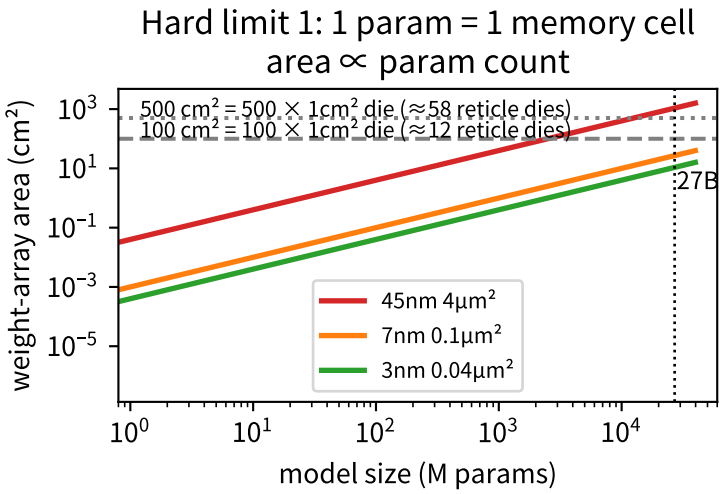
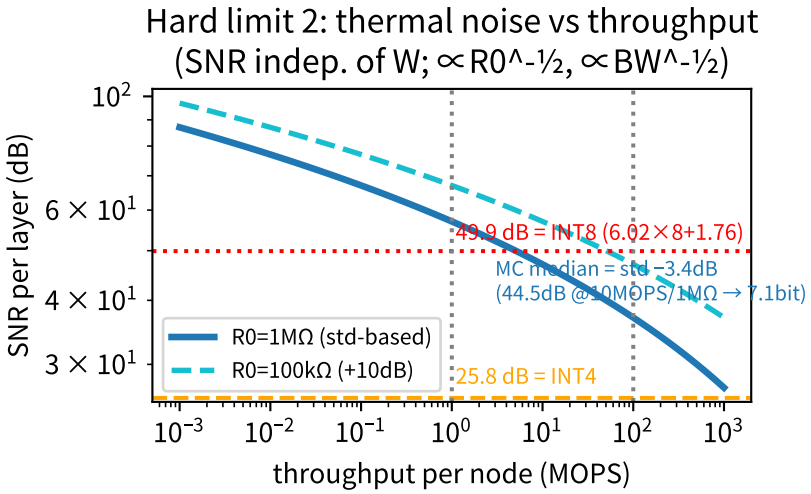
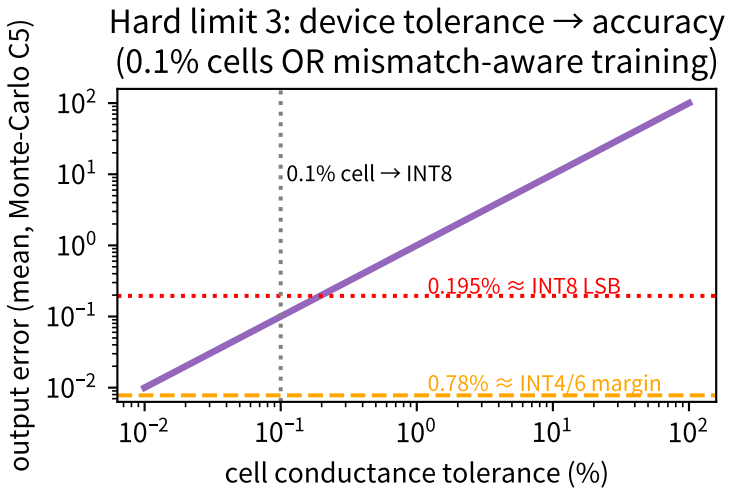

Hard limit 1: 1 param = 1 memory cell
area ∝ param count
500 cm² = 500 × 1cm² die (≈58 reticle dies)
100 cm² = 100 × 1cm² die (≈12 reticle dies)
27B
45nm 4μm²
7nm 0.1μm²
3nm 0.04μm²
weight-array area (cm²)
model size (M params)


Hard limit 2: thermal noise vs throughput
(SNR indep. of W; ∝R0^-½, ∝BW^-½)
49.9 dB = INT8 (6.02×8+1.76)
MC median = std −3.4dB
(44.5dB @10MOPS/1MΩ → 7.1bit)
R0=1MΩ (std-based)
R0=100kΩ (+10dB)
25.8 dB = INT4
SNR per layer (dB)
throughput per node (MOPS)
1 MOPS
100 MOPS


Hard limit 3: device tolerance → accuracy
(0.1% cells OR mismatch-aware training)
0.1% cell → INT8
0.195% ≈ INT8 LSB
0.78% ≈ INT4/6 margin
output error (mean, Monte-Carlo C5)
cell conductance tolerance (%)